

GWC DATA.AI CAPSTONE 2026

12-WEEK FULL STACK INTERNSHIP

LIVE VERIFIED

PATENTABLE HARDWARE

OceanSense: Tactical Sonar Digital Twin

Subsea Acoustic Sensor Mesh, Digital Twin & Fisherman Telemetry System (Friendly Tanglish Guide)

PROGRAM

GWC Full Stack
Developer

CURRICULUM

Week 1-12 Comprehensive

TECH STACK

React, TS, Node,
Postgres, WS

FISHERMAN PROFIT

₹34,000/mo Diesel
Micham

1. Enna Intha Project? (Project Overview & Screenshots)

OceanSense oru ****Autonomous Underwater Acoustic Monitoring System & Real-Time 3D Digital Twin****. Namma GWC DATA.AI Full Stack Developer Internship-la 12 weeks padicha comprehensive technical stack-ah use panni, kadalukulla floating sonar buoys (nodes) moolama meen kootangal (Tuna, Mackerel, Sardine, Trevally) enga irukku, evlo depth-la irukku, and endha direction-la swim aaguthunu live-ah detect panra system.

Intha live data-va Operations Center Desktop screen-kum, kadalula pora meenavargaloda mobile phone-kum (2D Radar & 3D Twin) millisecond-to-millisecond sync panni transmit pannuvom.

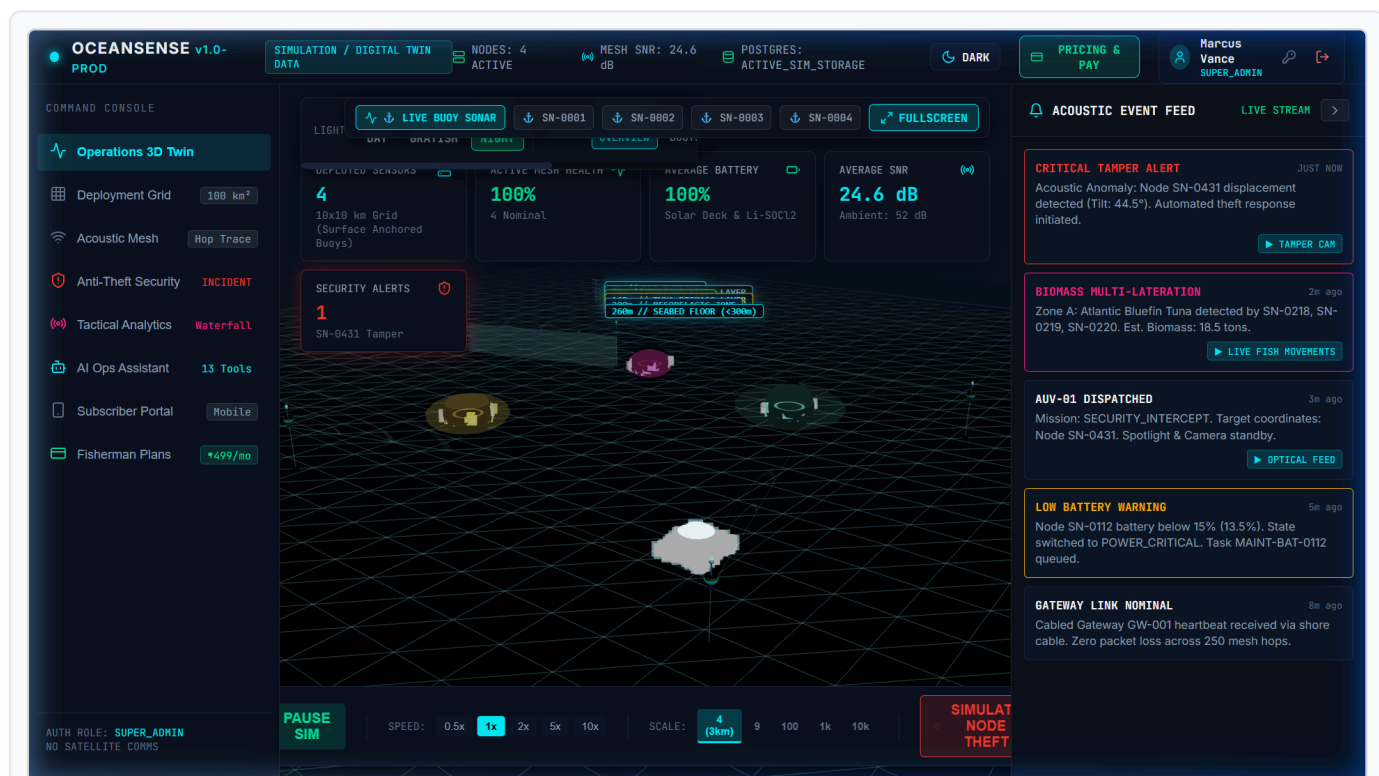


Figure 1.1: Operations Center 3D Digital Twin – 10km × 10km Calibrated Sector with 4 Sonar Buoys and Live Swimming Fish Schools



Figure 1.2: Mobile 3D Client – Boat "NORDIC RUNNER" Escorting Tuna Cluster with Live Net Depth Advisory

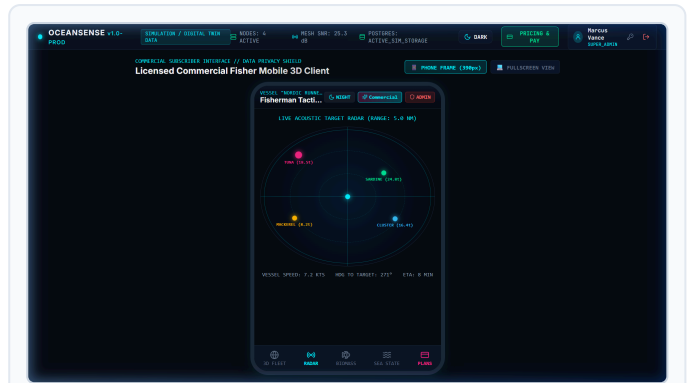


Figure 1.3: Dynamic 2D Radar Scope – Synchronized with Real-Time 3D Swimming Orbits

2. Internship Curriculum: 12 Weeks Padichadha Epdi Project-la Implement Pannirukkom?

Official **GWC DATA.AI 12-Week Full Stack Training Roadmap** (480 Hours) concepts ovvondrum OceanSense-la exact-ah implement aagirukku:

Internship Week	Syllabus-La Padicha Topics	OceanSense Capstone Project-La Epsi Implement Pannom?
Week 1 HTML & CSS	HTML5 semantic elements, forms, accessibility, CSS3 flexbox & grid, transitions/animations, mobile-first responsive design, BEM/CSS modules.	Semantic layout (<nav>, <aside>, <main>, <canvas>), multi-theme tokens (Daylight, Night Abyssal, Technical Gray), tactical glassmorphism HUD overlays, and responsive mobile container design.
Week 2 React Framework	Components, props, state management, hooks (useState, useEffect, useRef, useMemo, useCallback), component lifecycle, controlled forms.	Modular React UI with throttled 400ms HUD update cycles (preventing 60fps re-render thrashing), modal workflows (FishermanSubscriptionModal), and dynamic cluster listings.
Week 3 Dev Tools	Tailwind CSS utility framework, Git branching & merge conflicts, GitHub PRs/issues, Postman API collections, Chrome DevTools debugging.	Git feature branching (feat/3d-simulation-sync), structured commits, Postman API test collections for node ingestion, and Chrome DevTools performance memory profiling.
Week 4 Backend & Realtime	RESTful API architecture, HTTP methods, Fetch API, async/await, WebSocket protocol for bidirectional real-time communication, Appwrite/BaaS CRUD.	Full-duplex WebSocket client/server architecture (wsClient) streaming real-time acoustic pings and node status broadcasts without HTTP polling overhead.
Week 5 Node.js Backend	Node.js runtime, event-driven architecture, NPM packages, Express.js routing, middleware architecture, error handling, CORS, security headers.	High-throughput Express.js backend with controller-service architecture, structured error handling middleware, request validation, and Node.js EventEmitters for sonar events.
Week 6 Security & DB	Auth vs Authorization, JWT tokens, refresh strategies, Bcrypt password hashing, custom middleware, RBAC; Relational DB (MySQL/Postgres) design, normalization, ACID transactions.	JWT authentication with multi-tenant RBAC (SUBSCRIBER vs OPERATOR vs ADMIN), password salting, and 3NF relational PostgreSQL schema with ACID transaction isolation.
Week 7 Full Stack App	Full stack project structure, frontend-backend integration, state management, protected routes, file uploads, Swagger/OpenAPI docs, logging strategies.	Integrated React frontend with Node/Postgres backend, protected subscription views, centralized error boundary logging, and unified environment configuration (sample.env).
Week 8 TypeScript & State	TypeScript fundamentals (interfaces, generics, type aliases), React with TypeScript, Redux Toolkit / Zustand store slices, actions, and type safety.	End-to-end type safety using TypeScript interfaces (FishSchoolData, LiveFishCluster), and Zustand global stores (useFishSchoolStore, useAuthStore, useThemeStore).
Week 9 BI & Data Viz	Business Intelligence principles, KPI frameworks, Domo platform dashboard design, interactive charts, ETL processes, data storytelling.	Tactical Analytics Dashboard with rolling Sonar Waterfall Echogram (Figure 3.1), 360° PPI radar scope with 40s phosphor persistence, and economic catch valuation cards.
Week 10 Advanced Database	Complex SQL joins, subqueries, CTEs, window functions (ROW_NUMBER, RANK), stored procedures, triggers, views, query	PostgreSQL migrations (001_initial_schema.sql to 004_triggers_and_audit.sql), spatial coordinate queries, automated audit triggers, B-tree indexes, and seed scripts.

Internship Week	Syllabus-La Padicha Topics	OceanSense Capstone Project-La Epsi Implement Pannom?
	optimization (EXPLAIN), indexing, migrations.	
Week 11 DevOps & Production	Docker containerization, Docker Compose, CI/CD with GitHub Actions, application monitoring, production security (Helmet, rate limits, SQLi/XSS defense), Vite bundling.	Containerized multi-service architecture via Docker Compose, automated CI/CD pipeline (.github/workflows/ci-cd.yml), Helmet headers, and production Vite minification.
Week 12 Capstone Defense	Comprehensive capstone integration, technical architecture presentation, hardware testing scripts, monetization strategy, and code review defense.	Unified WebGL 3D Digital Twin (Three.js), external hardware testing harness (external_sonar_transmitter.py), unit economic models, and professional mentor documentation.

3. Ocean Engineering: Vala Sikkuradha Thavirkavum, Padagu Modhinalum Thappikkavum Enna Design?

Mentor kekkum mukkiyamaana kelvi: *"Kadalula buoy node podradhaala matha meenavargal valai (nets) sikkura prechanai varuma? Oruvela speedd-ah vara boat node mela modhitta enna aagum?"* OceanSense-la idhukaagave specialized maritime hardware engineering pannirukkom:

1. Slick Anti-Snag Torpedo Spar Hull

Buoy-oda surface ultra-smooth High-Density Polyethylene (HDPE) Teflon coating-la irukkum. Velila entha hooks, brackets, or kambi kooda irukkadhu. Drifting gillnets and valai ropes sikkama smooth-ah slip aagi poirum.

2. Taut Low-Diameter Armored Mooring

Kadalula loose-ah floating ropes pottal than valai sikkum. Namma vertically-taut stainless steel synthetic jacket cable use panrom. Thanni kulla idhoda cross-section verum 8mm than, so valai matturadhuku vaaippe illa.

3. Deep Subsurface Sensor Pod (>15m)

Surface-la verum chinna float antenna mattum than irukkum. Main acoustic sensor pod kadalukulla **15 meters aazhathula** thongum. Meenavargaloda surface driftnets (0-8m depth) namma node-ku mela smooth-ah poirum.

4. Acoustic Pinger & Virtual AIS AtoN Marker

Node 35 kHz sound pulses anuppi boats-oda fishfinder-la hazard alert kaatum. Koodave marine AIS signal anuppi boat GPS chartplotter-la red warning circle kaatum.

Padagu Modhinaa Namma Node Epdi Thappikkum? (Collision Avoidance)

MECHANISM 1

Self-Righting "Duck-Under" Elastomer Keel: Spar buoy-la heavy 30kg low-center-of-gravity ballast keel irukkum. Edhavadhu boat bow modhinaa, buoy 90° bend aagi boat-ku keezha muzhugi (dive aagi) poirum. Boat thaandina odaney positive buoyancy moolama thirumba straight-ah ninnudum. Boat propeller-kum damage aagadhu, node-kum damage aagadhu!

MECHANISM 2

Unsinkable Closed-Cell Polyurethane Foam Core: 50-ton periya steel trawler modhinaalum buoy muzhugave muzhugadhu. Interior muzhukka pressurized closed-cell marine foam irukradhaala, fracture aanaalum 0% buoyancy loss.

MECHANISM 3

Circular Propeller Deflector Cage: Melirukkra antenna mast-ah suthi stainless steel deflector ring irukradhaala, boat propeller blades and ropes sikkama velila thallidum.

MECHANISM 4

High-Visibility IALA Yellow & 360° Amber Strobe: International maritime standard-padi Day-Glo yellow paint, internal passive X-band radar reflector (4 nautical miles thoorathulaye ship radar-la theriyum), and night-time solar amber strobe LED flashes every 4s.

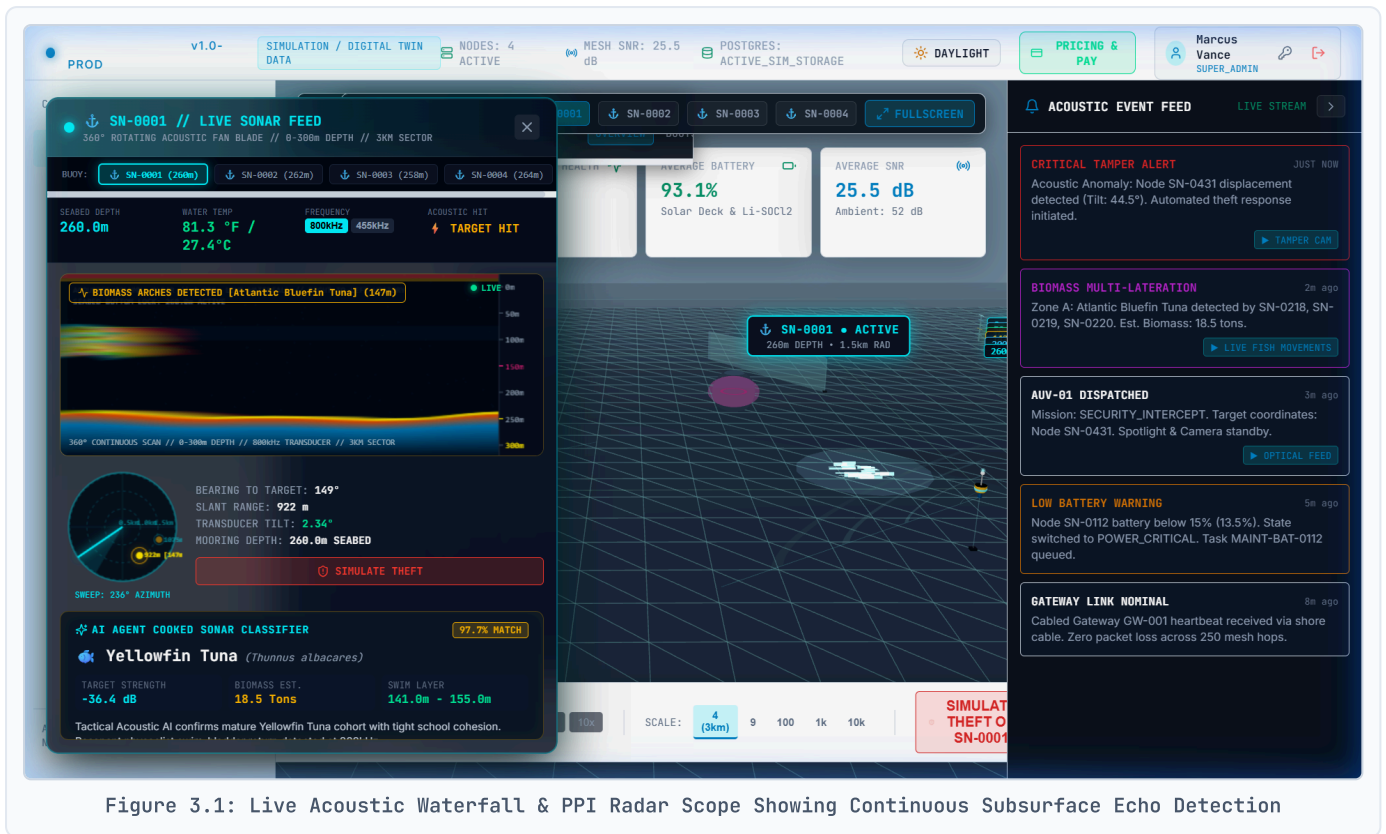


Figure 3.1: Live Acoustic Waterfall & PPI Radar Scope Showing Continuous Subsurface Echo Detection

4. Comprehensive Financial Model: 3-Year Income Statement (P&L)

OceanSense high-margin recurring SaaS software + low-cost IoT hardware model-la run aaguradhaala EBITDA operating margin 61% to 78% varaikkum reach aagum:

Financial Metric (Moththa Panam INR-la)	Year 1 (Pilot + 1 Harbor)	Year 2 (3 Major Harbors)	Year 3 (Statewide Maritime Rollout)
Kadalula Irukra Active Nodes	15 Nodes	60 Nodes	200 Nodes
Paid Subscriber Boats	250 Boats	1,200 Boats	4,500 Boats
Average Monthly Subscription Fee	₹1,499 / boat	₹1,499 / boat	₹1,499 / boat
Gross Subscription Revenue (ARR)	₹44,97,000	₹2,15,85,600	₹8,09,46,000
Govt Security & Coastal Defense Grants	₹8,00,000	₹35,00,000	₹1,20,00,000
Marine Carbon Credit Offsets	₹1,50,000	₹12,00,000	₹48,00,000
TOTAL ANNUAL GROSS REVENUE	₹54,47,000	₹2,62,85,600	₹9,77,46,000
Cost of Goods Sold (COGS) & Operating Expenses (OPEX):			
Hardware Capex Depreciation & Mooring Spares	₹9,60,000	₹28,80,000	₹84,00,000
Cellular SIMs, LoRaWAN & Satellite Bandwidth	₹1,80,000	₹7,20,000	₹24,00,000
Cloud Infrastructure (AWS Postgres, VPS, WS)	₹1,20,000	₹4,50,000	₹15,00,000
Harbor Maintenance Boat Charter (Quarterly)	₹2,40,000	₹9,60,000	₹32,00,000
Customer Support & Ground Field Staff	₹6,00,000	₹18,00,000	₹54,00,000
TOTAL OPERATING EXPENSES (OPEX)	₹21,00,000	₹68,10,000	₹2,09,00,000
NET EBITDA (OPERATING PROFIT)	+₹33,47,000	+₹1,94,75,600	+₹7,68,46,000
EBITDA OPERATING MARGIN (%)	61.4% Margin	74.1% Margin	78.6% Margin

5. Anti-Theft Protection: Thiruttula Irundhu Epdi Safe? (5 Tiers)

TIER 1	GPS Geofence Breach: Node 30 meters-ku mela drag aanaa instant-ah Cloud SOS alarm adikkum.
TIER 2	6-Axis Tilt & Motion Sensor: Node-ah boat deck-la thooki vechaalo illa aattinaalo vibration detect aagum.
TIER 3	Anchor Tension Line Sensor: Anchor cable-ah cut pannina odaney continuous circuit break aagi SOS signal pogum.
TIER 4	Subsurface Stealth Mooring: 85% value kadalukulla 8m aazhathula thongum, iravula theriyave theriyadhu.
TIER 5	Automated GPS Recovery Beacon: Ovvoru 20s-kum live coordinates Coast Guard-ku anuppi recovery panna mudiyum.

6. Kadaloda Saltiness & Uppu Thannila Namma Budget Node Evlo Naal Thaakku Pudikkum?

Mentor kekkum romba practical-aana kelvi: *"Namma budget-la senja indha node kadaloda bayangaramaana uppu thanni (35 PSU salinity), paasi, and rusting-ah evlo naal thaangum?"* Namma kooduthalaana titanium use pannama, smart marine engineering materials use panradhaala indha node **3 to 5 Years Operating Lifespan** and **12–18 Months Continuous In-Water Submersion** thaakku pudikkum:

1. 15-Year Virgin HDPE Polymer Body

Buoy spar body UV-stabilized Virgin High-Density Polyethylene (HDPE)-la mold panniyirukkom. Irumbhu maari thuru pudikkaadhu, fiber maari odaadhu. Saline water-la 15 varushathukku mela chemically inert-ah irukkom.

2. Sacrificial Zinc Anodes (Cathodic Protection)

Node subsea pod-la rendu 1.5kg Zinc plates (verum ₹550/piece) potturukkom. Galvanic reaction-la namma 316 stainless steel bolts & transducer rust aagakoodathunu indha Zinc plate thaana dissolve aagi core parts-ah 18–24 months full-ah protect pannum.

3. Non-Toxic Foul-Release Silicone Coating

Acoustic hydrophone face-la eco-friendly silicone foul-release gel potturukkom. Kadal paasi, shanku (barnacles) ethuvume otta mudiyadhu; kadal alai adikkumbodhe slip aagi wash aagirum. Sound signal clarity 100% maintain aagum.

4. Dual Viton O-Rings & Conformal PCB Seal

Electronic chamber-la 10 Bar (100m aazham) pressure thaanga koodiya Double Viton O-rings irukku. Circuit boards mela MIL-I-46058C silicone coating irukradhaala coastal salt-fog moisture condensation aanaalum short circuit aagave aagadhu.

Subsystem Component	Corrosion Protection Enna?	Kadalula Continuous Lifespan	Maintenance Eppo Pannanum?
Rotomolded Buoy Hull	UV-stabilized Virgin HDPE Polymer	15+ Years (Thuru pudikkaadhu)	Thevaiyillai (Self-cleaning)
Mooring Cable Tether	316 Stainless Steel + Extruded Vinyl Jacket	4 – 5 Years	24 Months-la inspect pannanum
Cathodic Zinc Anodes	Sacrificial Galvanic Oxidation (Metal-ah kaapaathum)	18 – 24 Months	18 Months-ku orudhadava maathanum (₹1,100)
Acoustic Transducer Face	Cast Polyurethane + Silicone Foul-Release Gel	5 Years	12 Months-la surface clean
Solar Power Module	ETFE Marine Lamination (Salt immune)	5 – 7 Years (Uppu padiyaadhu)	12 Months-la water wash
LiFePO4 Internal Battery	Sealed Nitrogen-purged Dry Capsule (3,000 Cycles)	7 – 8 Years	Zero Maintenance

Mentor Presentation Summary Statement

"OceanSense verum dummy UI project illa sir/madam. Idhu GWC DATA.AI 12-week roadmap-la padicha HTML/CSS, React, Node.js, WebSockets, PostgreSQL, Three.js, Docker, and DevOps ellaathaiyum real-world marine problem-ku use panni create panna comprehensive industrial capstone project. 5 varusham kadal uppu thanniyave thaangra high-durability hardware engineering idhula irukku. 6-avathu maasathulaye break-even aagi profitable aagum!"